

Linux Commands Cheatsheet

Basic File Operations

ls -lh file	Display file permissions, size, owner etc
touch file	Creates an empty file
cp file1 file2	Copy file1 to file2. File2 can be a directory
mv file dir	Move a file to a directory
mv file1 file2	Rename file1 to file2
rm file	Delete a file
ls -lah	List all the contents in a directory
mkdir data	Creates a directory data
cp -r dir1 dir2	Copy dir1 and its contents to dir2
rm -rf dir	Delete a directory and its contents
pwd	Print current working directory
stat file	Display attributes of files and directories
wc file	Count bytes, words, lines in a file or STDIN
file file	Identify (guess) the type of a file.
type cd	Find out whether cd binary is built-in, alias or external binary file

File Viewing

cat file.txt	Print the contents of a text file
tac file.txt	Prints text in reverse
more file.txt	View large text files one page at a time
less data.txt	Same as more but with more features
head -n 5	View the first 5 line of a text file
tail -n 5	View the last 5 line of a text file
nl file	View text files with their lines numbered
strings file	Display text that's embedded in a binary file

Print Text

echo "Hello World"	Print Hello World on the standard output
printf "%5d\n" 42	Print formatted text on standard output
yes "Hello World"	Print repeated text on the standard output
seq 1 5	Print a sequence of numbers from 1 to 5
clear	Clear the terminal screen or window

File Search

locate file	Searches for files and directories
which cd	Searches the location of the cd binary
whereis ls	Find ls' binary docs, and source files
find /data -name hello.txt	Searches for "hello.txt" in the /data directory

Directory Traversal

cd or cd ~	Navigate to the user's home directory
cd ..	Navigate to the parent directory
cd -	Switch to the previous working directory
cd /	Navigate to the root directory
cd /tmp	Changes the current directory to /tmp

Disk Management

df -h	Report file system disk space usage
du -h /home	Estimate file space usage in the home dir
fdisk -l	List available partitions on a disk
cfdisk	Create partitions
lsblk	List block devices
mount /dev/sda /mnt	Mount /dev/sda partition to /mnt directory
umount /mnt	Unmount the mounted partition in /mnt dir
findmnt	Displays if about all mounted filesystems
fscck /dev/sda	Check a disk partition for errors

Secure Shell (SSH)

ssh traw@10.1.3.1	Remote login to 10.1.3.1
ssh-agent -t rsa	Generate SSH rsa key pair
ssh-copy-id	Copy ssh public key to a remote host
sshpass	Non-interactive ssh password auth tool

File Permissions

chmod +x	Set execute permissions to a file
chmod u+s script.sh	Set SUID permissions to a file
chmod g+s dir	Set SGID permissions to a directory
chmod +t dir	Set Sticky Bit permissions to a directory
chgrp devops file.txt	Changes file.txt group owner to devops
chmod 644 script.sh	Set the file perms to be read/write for the owner, and read-only for group and others
chown traw:sys file	Changes file owner to traw and group owner to sys
umask 022	Sets the default perms for newly created files to 644 and for directories to 755

History

sudo !!	Execute the previous command with sudo
^cat^tac	Replace previous cat command with tac
history	Display command line history
!\$	Last argument of the previous command
!50	Execute the 50th command in history

Process Management

ps	Display a snapshot of running processes
ps aux	Display all processes of all users
top	Display real-time view of running processes
pgrep firefox	Find the process ID of firefox
pidof firefox	Find the process ID of firefox
kill 6732	Terminate a process with PID of 6732
killall proc	Kills all processes named 'proc'
pkill firefox	Terminates the firefox process
bg	Resumes suspended jobs in the background
fg	Brings a suspended job to foreground
jobs	List active jobs in the current shell
renice 12 PID	Changes priority of process with given PID
pstree	Displays a tree of running processes

Networking

ping sysxplore.com	Sends ICMP packets to sysxplore.com
ip addr	Displays all network interfaces information
ifconfig	Shows network interfaces configuration
whois sysxplore.com	Displays domain's registration information
route	Display the routing table
ss	Display information about network sockets
netstat	Displays network information and statistics
dig sysxplore.com	Queries DNS, provides domain's DNS info
wget -curl	Download file from the specified url
curl sysxplore.com	Retrieves sysxplore.com home page

Compression/Archives

tar -cf backup.tar /home	Creates a tar archive of /home dir
tar -xf backup.tar	Extract files from "backup.tar" archive
tar -czvf data.tar.gz /home	Creates compressed archive of /home
gunzip data.gz	Uncompress data.gz file
zip -r data	Zip the data directory
unzip data.zip	Unzip the data.zip file
gzip data	Compresses "data" into "data.gz", original is removed

User Group Management

groups	Print the group membership of a user
groupadd devops	Create a new group called devops
groupdel devops	Deletes the devops group
groupmod -n sysops sys	Changes sysops group name to sys

User Management

id	Displays the user's UID, GID, and groups
whoami	Displays who is currently logged in
finger traw	Print information about user traw
useradd -u 1002 traw	Creates a new user traw with a specific UID
userdel traw	Deletes the user account named traw
chfn traw	Change a user's personal information
usermod -aG sudo traw	add user traw to the sudo group
gpasswd -a traw sudo	add user traw to the sudo group
gpasswd -d traw sudo	Remove user traw from the sudo group
passwd	Change user password
passwd traw	Change user traw's password
chsh -s /bin/zsh	Change user shell to zsh
su james	Switch to user james

Access Control Lists

getfacl file	Display ACL permissions of a file or directory
setfacl -m u:traw:r-x file	Set read/execute ACL perms for the user traw
setfacl -m g:sysops:r-w file	Set read/write ACL perms for the group sysops
setfacl -x u:traw file	Remove user traw ACL permissions
setfacl -x u:devops file	Remove the group devops ACL perms
setfacl -b file	Remove all ACL perms and keep default file permissions

File Transfer

scp file.txt user@rhost:/remote/dir	Copies file.txt to remote host's specified directory
rsync -a ~/ubuntu /backup/	Synchronizes content from source directory to destination directory, preserving attributes
rsync -a /var/www/web/ user@rhost:/data/backup/	Synchronizes local directory to remote, preserving attributes

Text Manipulation

grep "linux" file.txt	Search for the word Linux in file.txt
tr "a-z" "A-Z" <file	Translate lowercase chars to uppercase
rev < file.txt	Print file.txt contents in reverse
sort file.txt	Sort lines of text by various criteria
uniq file.txt	Print only unique lines in file.txt
vimdiff file1 file2	Line-by-line comparison of two files in vim
diff file1 file2	Comparison of two files on terminal
awk 'print \$1' file.txt	Print the first column of the file.txt
sed 's/cat/bat/g' file	Substitute all cat occurrence with bat in file

Linux Commands From A-Z

A

- adduser** Used to add user to the system
- addgroup** Used to add group to the system
- addr** Zline convert addresses into file names and line numbers
- alias** Create an alias for Linux commands
- anacron** Used to run commands periodically.
- apropos** Search the manual pages.
- apt** Search and install software packages (Debian/Ubuntu)
- apt-get** Search and install software packages (Debian/Ubuntu)
- arp** Display and modify ARP table.
- ar** A utility to create/modify/extract from archives.
- aspell** Interactive spell checker.
- as** A portable GNU assembler.
- at** Schedule a command to run once at a particular time
- awk** Find and Replace text, database sort/validate/index

B

- basename** Strip directory and suffix from filenames
- base32** Base32 encode/decode data and print to standard output
- base64** Base64 encode/decode data and print to standard output
- bash** GNU Bourne-Again Shell
- bc** Access the GNU bc calculator utility.
- bg** Send processes to the background.
- bind** Used to attach a name to a socket.
- break** Used to break from a loop.
- builtin** Run a shell builtin
- bzip2** Compress or decompress named file(s)
- bzcmp** Used to call the cmp program for bzip2 compressed files.
- bzdiff** Used to call the diff program for bzip2 compressed files.
- bzgrep** Used to call grep for bzip2 compressed files.
- bzip2** A block-sorting file compressor used to shrink given files.
- bzless** Used to apply 'less' to bzip2 compressed files.
- bzmore** Used to apply 'more' to bzip2 compressed files.

C

- cal** Show calendar
- caller** Return the context of any active subroutine call
- case** Conditionally execute a command by matching a pattern.
- cat** Concatenate and print (display) the content of files
- cc** GNU C and C++ compiler.
- cd** Change Directory
- cdisk** Partition table manipulator for Linux
- chattr** Change file attributes on a Linux file system
- chgrp** Change group ownership
- chmod** Change access permissions
- chown** Change file owner and group
- chpasswd** Update passwords in batch mode
- chfn** Used to change real user name and information.
- chroot** Run a command with a different root directory
- chsh** Switch login shell.
- chkconfig** System services (runlevel)
- cksum** Print CRC checksum and byte counts
- clear** Clear terminal screen
- cmp** Compare two files
- comm** Compare two sorted files line by line
- command** Run a command - ignoring shell functions
- compress** Used to compress one or more file(s).
- continue** Resume the next iteration of a loop
- cp** Copy one or more files to another location
- cpio** Copy files to and from archives
- cron** Daemon to execute scheduled commands
- crontab** Schedule a command to run at a later time
- csplit** Split a file into context-determined pieces
- curl** Transfer data from or to a server
- cut** Divide a file into several parts

D

- date** Display or change the date & time

- dd** Data Duplicator - convert and copy a file, write disk headers, boot records
- ddrescue** Data recovery tool
- declare** Declare variables and give them attributes
- df** Display free disk space
- diff** Display the differences between two files
- diff3** Show differences among three files
- dig** DNS lookup
- dif** Briefly list directory contents
- dircolors** Colour setup for 'ls'
- dirname** Convert a full pathname to just a path
- dirs** Display list of remembered directories
- dos2unix** Windows/MAC to UNIX text file format converter
- dmesg** Print kernel & driver messages
- dpkg** Package manager (Debian/Ubuntu).
- du** Estimate file space usage

E

- echo** Display message on screen
- egrep** Search file(s) for lines that match an extended expression
- eject** Eject removable media
- enable** Enable and disable builtin shell commands
- env** Environment variables
- ethtool** Ethernet card settings
- eval** Evaluate several commands/arguments
- exec** Execute a command
- exit** Exit the shell
- expand** Convert tabs to spaces
- export** Set an environment variable
- expr** Evaluate expressions

F

- false** Do nothing, unsuccessfully
- fdformat** Low-level format a floppy disk
- fdisk** Partition table manipulator for Linux
- fg** Send job to foreground
- fgrep** Search file(s) for lines that match a fixed string
- file** Determine file type
- find** Search for files that meet a desired criteria
- fmt** Reformat paragraph text
- fold** Wrap text to fit a specified width
- for** Expand words, and execute commands
- format** Format disks or tapes
- free** Display memory usage
- fsck** File system consistency check and repair
- ftp** File Transfer Protocol
- function** Define Function Macros
- fuser** Identify/kill the process that is accessing a file

G

- gawk** Find and Replace text within file(s)
- getopts** Parse positional parameters
- grep** Search file(s) for lines that match a given pattern
- groupadd** Add a user security group
- groupdel** Delete a group
- groupmod** Modify a group
- groups** Print group names a user is in
- gzip** Compress or decompress named file(s)

H

- hash** Remember the full pathname of a name argument
- head** Output the first part of file(s)
- help** Display help for a built-in command
- history** Command History
- hostname** Print or set system name
- htop** Interactive process viewer

I

- iconv** Convert the character set of a file
- id** Print user and group id's

- pathchk** Check file name portability
- Perf** Performance analysis tools for Linux
- ping** Test a network connection
- pgrep** List processes by name
- kill** Kill processes by name
- popd** Restore the previous value of the current directory
- pr** Prepare files for printing
- printcap** Printer capability database
- printenv** Print environment variables
- printf** Format and print data
- ps** Process status
- pushd** Save and then change the current directory
- pv** Monitor the progress of data through a pipe
- pwd** Print Working Directory

Q

- quota** Display disk usage and limits
- quotacheck** Scan a file system for disk usage

R

- ram** ram disk device
- rar** Archive files with compression
- rcp** Copy files between two machines
- read** Read a line from standard input
- readarray** Read from stdin into an array variable
- readonly** Mark variables/functions as readonly
- reboot** Reboot the system
- rename** Rename files
- renice** Alter priority of running processes
- rm** Remove files
- rmrmdir** Remove folder(s)
- rsync** Remote file copy (Synchronize file trees)

S

- screen** Terminal Multiplexer
- scp** Secure copy (remote file copy)
- sdiff** Merge two files interactively
- sed** Stream Editor
- select** Accept user choices via keyboard input
- seq** Print numeric sequences
- set** Manipulate shell variables and functions
- setp** Secure File Transfer Program
- shift** Shift positional parameters
- shopt** Shell Options
- shuf** Generate random permutations
- shutdown** Shutdown or restart linux
- sleep** Delay for a specified time
- slocate** Find files
- sort** Sort text files
- source** Run commands from a file
- split** Split a file into fixed-size pieces
- ss** Socket Statistics
- ssh** Secure Shell client (remote login program)
- stat** Display file or file system status
- strace** Trace system calls and signals
- su** Substitute user identity
- sudo** Execute a command as another user
- sum** Print a checksum for a file
- suspend** Suspend execution of this shell
- sync** Synchronize data on disk with memory

L

- lconvi** Convert the character set of a file
- ld** Print user and group id's

- ifup** Start a network interface up
- import** Capture an X server screen and save the image to file
- install** Copy files and set attributes
- iotstat** Report CPU and i/o statistics
- ip** Routing, devices and tunnels

- jobs** List active jobs
- join** Join lines on a common field
- K**
- kill** Kill a process by specifying its PID
- killall** Kill processes by name

L

- less** Display output one screen at a time
- let** Perform arithmetic on shell variables
- link** Create a link to a file
- ln** Create a symbolic link to a file
- local** Create a function variable
- locate** Find files
- logname** Print current login name
- logout** Exit a login shell
- look** Display lines beginning with a given string
- lpc** Line printer control program
- lp** Print files
- lprint** Print a file
- lprintd** Delete a print job
- lprintq** List the print queue
- lprm** Remove jobs from the print queue
- lsattr** List file attributes on a Linux second extended file system
- lsblk** List block devices
- ls** List information about file(s)
- lsotf** List open files
- lspci** List all PCI devices

M

- make** Recompile a group of programs
- man** Help manual
- mapfile** Read lines from standard input into an indexed array variable
- mkdir** Create new folder(s)
- mkfifo** Make FIFOs (named pipes)
- mkfile** Make a file
- mkisofs** Create a hybrid ISO9660/JOLIE/HFS filesystem
- mklnod** Make block or character special files
- mktemp** Make a temporary file
- more** Display output one screen at a time
- most** Browse or page through a text file
- mount** Mount a file system
- mttools** Manipulate MS-DOS files
- mtr** Network diagnostics (traceroute/ping)
- mv** Move or rename files or directories
- mmv** Mass Move and rename (files)

N

- nc** Netcat, read and write data across networks
- netstat** Networking connections/stats
- nft** Netfilter for packet filtering and classification
- nice** Set the priority of a command or job
- nl** Number lines and write files
- nohup** Run a command immune to hangups
- notify-send** Send desktop notifications
- nslookup** Query internet name servers interactively

O

- open** Open a file in its default application

T

- tail** Output the last part of a file
- tar** Store, list or extract files in an archive
- tee** Redirect output to multiple files
- test** Evaluate a conditional expression
- timeMeasure** Program running time
- timeout** Run a command with a time limit
- times** User and system times
- tmux** Terminal multiplexer
- touch** Change file timestamps
- top** List processes running on the system
- tpset** Set terminal-dependent capabilities, color, position
- traceroute** Trace Route to Host
- trap** Execute a command when the shell receives a signal
- tr** Translate, squeeze, and/or delete characters
- true** Do nothing, successfully
- tsort** Topological sort
- tty** Print filename of terminal on stdin
- type** Describe a command

U

- ulimit** Limit user resources
- umask** Users file creation mask
- umount** Unmount a device
- unalias** Remove an alias
- uname** Print system information
- unexpand** Convert spaces to tabs
- uniq** Uniquely files
- units** Convert units from one scale to another
- unix2dos** UNIX to Windows or MAC text file format converter
- unrar** Extract files from a rar archive
- unset** Remove variable or function names
- unshar** Unpack shell archive scripts
- until** Execute commands (until error)
- uptime** Show uptime
- useradd** Create new user account
- userdel** Delete a user account
- usermod** Modify user account
- users** List users currently logged in
- uuecode** Encode a binary file
- uudecode** Decode a file created by uuecode

V

- v** Verbosely list directory contents ('ls -l -b')
- vdif** Verbosely list directory contents ('ls -l -b')
- vi** Text Editor
- vmstat** Report virtual memory statistics

W

- w** Show who is logged on and what they are doing
- wait** Wait for a process to complete
- watch** Execute/display a program periodically
- wc** Print byte, word, and line counts
- whereis** Search the user's \$path, man pages and source files for a program
- which** Search the user's \$path for a program file
- while** Execute commands
- who** Print all usernames currently logged in
- whoami** Print the current user id and name ('id -un')
- wget** Retrieve web pages or files via HTTP, HTTPS or FTP
- write** Send a message to another user

XYZ

- xargs** Execute utility, passing constructed argument list(s)
- xdg-open** Open a file or URL in the user's preferred application.
- xxd** Make a hexdump or do the reverse
- xz** Compress or decompress .xz and .lzma files
- yes** Print a string until interrupted